LECTURE 10: SEMIPARAMETRIC EFFICIENCY BOUNDS

Guido Imbens – Stanford University

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OUTLINE

TERMINOLOGY

- Score function $\mathcal{S}(\cdot;\theta) = \frac{\partial \ln f}{\partial \theta}(z;\theta)$
- Information Matrix $\mathcal{I}(\theta) = -\mathbb{E}\left[\frac{\partial^2 \ln f}{\partial\theta\partial\theta^\top}(z;\theta)\right]$
- Cramér–Rao variance bound
- Efficient Score function $\psi(\cdot)$
- Influence function $\varphi(\cdot)$
- Asymptotic Linear Representation of estimator $\hat{\theta} = \theta + \frac{1}{N}\sum_{i=1}^{N} \varphi(Z_i\theta) + o_p(N^{-1/2})$
- semi-parametric efficiency bound
- Riesz representor $\phi(\cdot)$
- Tangent set

INTRODUCTION

- We are interested in estimating the average adjusted difference

$$\tau \equiv \mathbb{E}\left[\mathbb{E}\left[Y_i|W_i = 1, X_i\right] - \mathbb{E}\left[Y_i|W_i = 0, X_i\right]\right]$$

- Under unconfoundedness this is interesting (equal to average treatment effect)
- How well can we estimate τ without functional form assumptions?
- Starting point:
  - We are interested in a scalar (in general, in a finite-dimensional parameter).
  - In fully parametric models we would be able to estimate these at $\sqrt{N}$ rate,

$$\sqrt{N}(\hat{\theta} - \theta^*) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\theta)^{-1})$$

  - But we dont have a fully parametric model!
  - semi-parametric model, with parametric component (estimand) τ, and non-parametric components (distributions). What can we say here?

INTRODUCTION

- Classification of Estimation Problems

$$Y \sim f_Y(y)$$

  - parametric:

$$f(y) = f(y;\theta), \quad \theta \text{ finite dimensional, unknown, } f(\cdot) \text{ known}$$

  - non-parametric

$$f(y) \text{ unknown, beyond smoothness restrictions}$$

  - semi-parametric (today!)

$$f(y) = f(y;\theta,h(\cdot)), \quad \theta \text{ finite dimensional, unknown, } f(\cdot) \text{ known, } h(\cdot) \text{ unknow}$$

INTRODUCTION

- Simple example of semiparametric problem
- Suppose $X_1, \ldots, X_N$ are independent and identically distributed.
- We are interested in estimating

$$\theta = \mathrm{pr}(X_i > 0)$$

- A natural estimator is

$$\hat{\theta} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}_{X_i > 0}$$

- Can we do better?
- No (but how can we be sure of that?)

MORE COMPLICATED QUESTIONS

- Suppose $(Y_i(\mathrm{C}), Y_i(\mathrm{T}), W_i, X_i)$ are iid, we observe $(Y_i(W_i), W_i, X_i)$, we assume

$$W_i \perp\!\!\!\perp \Big(Y_i(\mathrm{C}), Y_i(\mathrm{T})\Big) \Big| X_i \qquad \text{(unconfoundedness)}$$

- We are interested in

$$\tau \equiv \mathbb{E}[Y_i(\mathrm{T}) - Y_i(\mathrm{C})] = \mathbb{E}\Big[\mathbb{E}[Y_i|X_i, W_i = 1] - \mathbb{E}[Y_i|X_i, W_i = 0]\Big]$$

- What is the best we can do?
- Answer: The best we can do is find an asymptotically linear estimator

$$\hat{\tau} = \theta + \frac{1}{N}\sum_{i=1}^{N} \varphi(Y_i, W_i, X_i) + o_p(N^{-1/2})$$

with asymptotic variance $\qquad \mathbb{E}\left[\varphi(Y_i, W_i, X_i)^2\right] = \mathbb{E}\left[\frac{\sigma_{\mathrm{T}}^2(X_i)}{e(X_i)} + \frac{\sigma_{\mathrm{C}}^2(X_i)}{1 - e(X_i)} + (\tau(X_i) - \tau)^2\right]$

- How does this change if we know $e(\cdot)$, or if we know $Y_i(\mathrm{T}) - Y_i(\mathrm{C})$ is constant?

BACK TO FIRST EXAMPLE

- We are interested in

$$\theta = \text{pr}(X_i > 0)$$

- Suppose we also know

$$X_i \sim \mathcal{N}(\mu, 1),$$

- Then we can do better than estimating θ with the fraction of positive observations, $\hat{\theta} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}_{X_i>0}$.
- First, estimate μ,

$$\hat{\mu} = \overline{X}$$

- Write θ as a function of μ, leading to

$$\hat{\theta} = 1 - \Phi(-\hat{\mu})$$

is better estimator (lower asymptotic variance).

EFFICIENCY IN PARAMETRIC MODELS: THE CRAMÉR–RAO BOUND

- Let the probability density function of a random variable X be $f_X(x|\theta)$ for some $\theta_0 \in \Theta$.
- Let $\hat{\theta}(X)$ be an unbiased estimator for θ_0.
- Suppose the derivative $\partial/\partial\theta$ can be passed under the integral sign in $\int f(x|\theta)dx$ and $\int \theta(x) f(x|\theta)dx$.
- Suppose the Fisher information

$$\mathcal{I}(\theta) = -\mathbb{E}\left[\frac{\partial^2 \ln f}{\partial\theta\partial\theta^\top}(X|\theta)\right],$$

  is finite.
- Then we have the variance bound

$$\mathrm{Var}(\hat{\theta}(X)) \geq \mathcal{I}(\theta_0)^{-1}.$$

MORE ON THE CRAMÉR–RAO BOUND

- If $X_1, X_2, \ldots, X_N$ are iid with common density $f_X(x|\theta)$, the implied bound on the variance is

$$N \cdot \mathrm{Var}(\hat{\theta}(X_1, \ldots, X_N)) \geq \mathcal{I}(\theta_0)^{-1},$$

- Here $\mathcal{I}(\theta_0)$ is the single observation Fisher information,

$$\mathcal{I}(\theta_0) = -\mathbb{E}\left[\frac{\partial^2 \ln f}{\partial\theta\partial\theta'}(X_i|\theta_0)\right].$$

- At the same time the limiting distribution of the mle is, under conditions discussed before, equal to

$$\sqrt{N}(\hat{\theta}_{\mathrm{mle}} - \theta_0) \xrightarrow{d} \mathcal{N}\left(0, \mathcal{I}(\theta_0)^{-1}\right).$$

- Hence in large samples: the mle is approximately unbiased and approximately achieves the Cramér–Rao bound.

DEFINITION OF LARGE SAMPLE EFFICIENCY

- This motivates the following definition of large sample efficiency:
- Let $X_1, X_2, \ldots$ be iid random variables with common density $f_X(x|\theta_0)$.
- A sequence of estimators $\hat{\theta}_N$, a function of $X_1, X_2, \ldots, X_N$ such that

$$\sqrt{N}(\hat{\theta}_N - \theta_0) \xrightarrow{d} \mathcal{N}\left(0, \mathcal{I}(\theta_0)^{-1}\right),$$

whatever the true value $\theta_0 \in \Theta$ is, is said to be asymptotically efficient.

ASIDE: RULING OUT SUPER-EFFICIENT ESTIMATORS

- Rule out super-efficient estimators that do well in a particular part of the sample space but not in others.
- For example, suppose $X_1, X_2, \ldots$ are iid normal with mean θ and variance 1. The mle is $\overline{X}_N$ with normalized variance (variance of $\sqrt{N}(\overline{X} - \theta)$) equal to 1. Now consider the estimator

$$\hat{\theta} = \begin{cases} \overline{X}_N & \text{if } N^{1/4} \cdot |\overline{X}_N| \geq 1 \\ 0 & \text{if } N^{1/4} \cdot |\overline{X}_N| < 1. \end{cases}$$

- If $\theta = 0$, the normalized variance is zero. (If $\theta = 0$, $\sqrt{N}(\hat{\theta} - \theta) \sim \mathcal{N}(0, \sigma^2)$, so $N^{1/2}(\hat{\theta} - \theta) \to 0$
- Hence at zero $\hat{\theta}$ does better, and if $\theta \neq 0$ this estimator does equally well as the mle (in large samples, where $N^{1/4} \cdot |\overline{X}_N| \geq 1$).

BACK TO EXAMPLE

- Suppose $X_1, X_2, \ldots$ are iid with unknown density function $f(x)$. The density depends on some parameters of interest θ and an unknown function $h(\cdot)$:

$$X \sim f(x|\theta, h(\cdot)).$$

- For example, if

$$\theta = \mathbb{E}\left[\mathbf{1}_{X_i > 0}\right],$$

we could specify the unknown function $h(\cdot)$ to be

$$h(x) = \begin{pmatrix} h_1(x) \\ h_0(x) \end{pmatrix} = \begin{pmatrix} f(x) \cdot \mathbf{1}_{x>0} \Big/ \int_0^\infty f(z)dz \\ f(x) \cdot \mathbf{1}_{x\leq 0} \Big/ \int_{-\infty}^0 f(z)dz \end{pmatrix}.$$

- In that case the density as a function of the two unknown parameters: (*i*) θ (finite dimensional) and (*ii*) $h(\cdot)$ (infinite dimensional) is

$$f(x) = f(x|\theta, h(\cdot)) = \mathbf{1}_{x>0} \cdot \theta \cdot h_1(x) + \mathbf{1}_{x\leq 0} \cdot (1-\theta) \cdot h_0(x).$$

KEY TRICK: MAKE IT A PARAMETRIC PROBLEM

- Suppose we actually know $h(x)$.
- Then we have a fully parametric model with

$$f(x;\theta) = \mathbf{1}_{x>0} \cdot \theta \cdot h_1(x) + \mathbf{1}_{x\leq 0} \cdot (1-\theta) \cdot h_0(x).$$

- The score function is

$$\mathcal{S}(x|\theta) = \frac{\partial}{\partial\theta} \ln\, f(x;\theta) = \frac{\mathbf{1}_{x>0}}{\theta} - \frac{\mathbf{1}_{x\leq 0}}{1-\theta} = \frac{\mathbf{1}_{x>0} - \theta}{\theta(1-\theta)}.$$

- The variance of the score, equal to the Fisher information, is

$$\mathcal{I}(\theta_0) = -\mathbb{E}\left[\frac{\partial^2 \ln\, f}{\partial\theta\partial\theta^\top}(X_i|\theta)\right]_{\theta=\theta_0} = \mathbb{E}\left[\mathcal{S}(X|\theta_0)^2\right] = \frac{1}{\theta_0(1-\theta_0)},$$

- The Cramér–Rao variance bound is the inverse of the Fisher information, so equal to $\theta_0(1-\theta_0)$.

SEMIPARAMETRIC EFFICIENCY

- Now consider the estimator

$$\hat{\theta} = \sum_{i=1}^{N} \mathbf{1}_{X_i > 0}/N.$$

- This has mean θ_0 and variance $\theta_0(1 - \theta_0)/N$.
- The variance of $\hat{\theta}$ is equal to the Cramer-Rao bound if I give you additional information.
- Hence $\hat{\theta}$ is efficient and $\theta_0(1 - \theta_0)$ is the semi-parametric variance bound.

THE GENERAL CASE

- Now let us do this more systematically.
- Suppose we know $h(\cdot)$ up to a finite dimensional parameter vector γ so that

$$f(x|\theta,\gamma) = f(x|\theta,h(\gamma)).$$

- Hence we can calculate the Cramér–Rao bound for θ.
- Partition the information matrix for (θ,γ) and its inverse in

$$\mathcal{I}(\theta,\gamma) = \begin{pmatrix} \mathcal{I}_{\theta\theta^\top} & \mathcal{I}_{\theta\gamma^\top} \\ \mathcal{I}_{\gamma\theta^\top} & \mathcal{I}_{\gamma\gamma^\top} \end{pmatrix} \qquad \text{and } \mathcal{I}(\theta,\gamma)^{-1} = \begin{pmatrix} \mathcal{I}^{\theta\theta^\top} & \mathcal{I}^{\theta\gamma^\top} \\ \mathcal{I}^{\gamma\theta^\top} & \mathcal{I}^{\gamma\gamma^\top} \end{pmatrix}$$

- The Cramér–Rao bound implies that

$$\text{ASV}(\hat{\theta}) \geq \mathcal{I}^{\theta\theta^\top} = \left(\mathcal{I}_{\theta\theta^\top} - \mathcal{I}_{\theta\gamma^\top}\mathcal{I}^{-1}_{\gamma\gamma^\top}\mathcal{I}_{\gamma\theta^\top}\right)^{-1},$$

where $\text{ASV}(\hat{\theta})$ is the (normalized by sample size) asymptotic variance of $\hat{\theta}$.

THE GENERAL CASE

- Let $\beta = (\theta, \gamma)$ be the complete parameter vector. Define the score function

$$\mathcal{S}_\beta(x;\beta) = \begin{pmatrix} \mathcal{S}_\theta(x;\theta;\gamma) \\ \mathcal{S}_\gamma(x;\theta,\gamma) \end{pmatrix} = \begin{pmatrix} \frac{\partial \ln f}{\partial \theta}(x;\theta,\gamma) \\ \frac{\partial \ln f}{\partial \gamma}(x;\theta,\gamma) \end{pmatrix}.$$

- Then

$$\mathcal{I}(\theta,\gamma) = \begin{pmatrix} \mathcal{I}_{\theta\theta^\top} & \mathcal{I}_{\theta\gamma^\top} \\ \mathcal{I}_{\gamma\theta^\top} & \mathcal{I}_{\gamma\gamma^\top} \end{pmatrix}$$

$$= \mathbb{E}\left[\begin{pmatrix} \mathcal{S}_\theta(X_i;\theta,\gamma)\mathcal{S}_\theta^\top(X_i;\theta,\gamma) & \mathcal{S}_\theta(X_i;\theta,\gamma)\mathcal{S}_\gamma^\top(X_i;\theta,\gamma) \\ \mathcal{S}_\gamma(X_i;\theta,\gamma)\mathcal{S}_\theta^\top(X_i;\theta,\gamma) & \mathcal{S}_\gamma(X_i;\theta,\gamma)\mathcal{S}_\gamma^\top(X_i;\theta,\gamma) \end{pmatrix}\right].$$

THE GENERAL CASE

- Then we can write the inverse of the Cramér–Rao variance bound (the information bound) for θ as as

$$\left(\mathcal{I}^{\theta\theta^\top}\right)^{-1} = \mathcal{I}_{\theta\theta^\top} - \mathcal{I}_{\theta\gamma^\top}\mathcal{I}^{-1}_{\gamma\gamma^\top}\mathcal{I}_{\gamma\theta^\top} = \mathbb{E}\left[\psi(X_i)\psi(X_i)^\top\right]$$

- Here $\psi(X_i)$ is the residual from the projection of the score for the parameters of interest θ, $\mathcal{S}_\theta(X_i)$, on the score for the nuisance parameter γ, $\mathcal{S}_\gamma(X_i)$:

$$\psi(x) \equiv \mathcal{S}_\theta(x) - \mathbb{E}\left[\mathcal{S}_\gamma(X_i)\mathcal{S}_\theta^\top(X_i)\right]^\top \mathbb{E}\left[\mathcal{S}_\gamma(X_i)\mathcal{S}_\gamma^\top(X_i)\right]^{-1}\mathcal{S}_\gamma(x).$$

- $\mathcal{S}_\theta(x)$ is the score function, and $\psi(x)$ is the efficient score function where we partial out the component from the nuisance parameters.
- $\psi(X_i)$ must have a smaller variance than $\mathcal{S}_\theta(X_i)$.

THE GENERAL CASE

- The space of all possibly nuisance scores (the score functions corresponding to all possible parametric models) is the tangent set or tangent space.
- If we can find the projection of $\mathcal{S}_\theta(\cdot)$ on the tangent set, then the residual from that projection, $\psi(\cdot)$ is the efficient score.
- The influence function is

$$\varphi(x) \equiv \left(\mathbb{E}[\psi(X_i)\psi(X_i)^\top]\right)^{-1}\psi(x)$$

- The asymptotically linear approx of the maximum likelihood estimator is

$$\hat{\theta}_{\mathrm{mle}} = \theta + \frac{1}{N}\sum_{i=1}^{N}\varphi(X_i) + o_p(N^{-1/2})$$

- The semi-parametric variance bound is

$$\mathbb{E}\left[\varphi(X_i)\varphi(X_i)^\top\right] = \left(\mathbb{E}\left[\psi(X_i)\psi(X_i)^\top\right]\right)^{-1}$$

BACK TO EXAMPLE

- Let us consider this in the example discussed earlier, where

$$f(x;\theta) = \mathbf{1}_{x>0} \cdot \theta \cdot h_1(x) + \mathbf{1}_{x\leq 0} \cdot (1-\theta) \cdot h_0(x).$$

- Let us parametrize $h(\cdot)$ as $h_1(x;\gamma_1)$ and $h_0(x;\gamma_0)$. The score function is

$$\mathcal{S}_\beta(x) = \begin{pmatrix} \mathcal{S}_\theta(x) \\ \mathcal{S}_{\gamma_1}(x) \\ \mathcal{S}_{\gamma_0}(x) \end{pmatrix} = \begin{pmatrix} \frac{\partial \ln f}{\partial \theta}(x;\theta,\gamma) \\ \frac{\partial \ln f}{\partial \gamma_1}(x;\theta,\gamma) \\ \frac{\partial \ln f}{\partial \gamma_0}(x;\theta,\gamma) \end{pmatrix} = \begin{pmatrix} (\mathbf{1}_{x>0} - \theta)/(\theta \cdot (1-\theta)) \\ \mathcal{S}_1(x;\gamma_1) \\ \mathcal{S}_0(x;\gamma_0) \end{pmatrix}$$

- Here

$$\mathcal{S}_1(x;\gamma_1) = \mathbf{1}_{x>0} \cdot \frac{\partial \ln h_1}{\partial \gamma_1}(x;\gamma_1), \qquad \mathcal{S}_0(x;\gamma_1) = \mathbf{1}_{x\leq 0} \cdot \frac{\partial \ln h_0}{\partial \gamma_1}(x;\gamma_0),$$

BACK TO EXAMPLE

- For any parametrization the score functions $\mathcal{S}_1(x;\gamma_1)$ and $\mathcal{S}_0(x;\gamma_0)$ satisfy

$$\mathbb{E}[\mathcal{S}_1(X_i;\gamma_1)] = \mathbb{E}[\mathcal{S}_0(X_i;\gamma_0)] = 0.$$

- The tangent space is a set of functions $s(\cdot)$, specifically the set of all linear combinations of possible score functions for the nuisance functions:

$$\mathcal{T} = \Bigg\{ s_1(x) + s_0(x) : \int_0^\infty s_1(x)\, f(x)dx = 0, \int_{-\infty}^0 s_0(x)\, f(x)dx = 0,$$

$$s_1(x) = \mathbf{1}_{x>0} s_1(x), s_0(x) = \mathbf{1}_{x\leq 0} s_0(x) \Bigg\}$$

BACK TO EXAMPLE

- Now, to consider the projection, let us first look at the correlation between elements of the tangent space, say $s(x)$, and the score for the parameter of interest,

$$\mathcal{S}_\theta(x) = (\mathbf{1}_{x>0} - \theta)/(\theta \cdot (1-\theta))$$

- This correlation turns out to be zero, so the residual from the projection of the score on the tangent set is equal to the score itself.
- To see the correlation is zero, note

$$\mathbb{E}\left[s(X_i) \cdot \mathcal{S}_\theta(X_i)\right] = \mathbb{E}\left[s_1(X_i) \cdot \mathcal{S}_\theta(X_i)\right] + \mathbb{E}\left[s_0(X_i) \cdot \mathcal{S}_\theta(X_i)\right]$$

$$= \mathbb{E}\left[s_1(X_i) \cdot (\mathbf{1}_{X_i>0} - \theta)/(\theta \cdot (1-\theta))\right] + \mathbb{E}\left[s_0(X_i) \cdot (\mathbf{1}_{X_i>0} - \theta)/(\theta \cdot (1-\theta))\right]$$

$$= \frac{1}{\theta(1-\theta)}\left\{\mathbb{E}\left[s_1(X_i) \cdot \mathbf{1}_{X_i>0}\right] + \mathbb{E}\left[s_1(X_i) \cdot \theta\right]\right\}$$

$$+ \frac{1}{\theta(1-\theta)}\left\{\mathbb{E}\left[s_0(X_i) \cdot \mathbf{1}_{X_i>0}\right] + \mathbb{E}\left[s_0(X_i) \cdot \theta\right]\right\}.$$

BACK TO EXAMPLE

- Consider the first term. By the definition of the tangent set,

$$\mathbb{E}\left[s_1(X_i)\cdot \mathbf{1}_{X_i>0}\right] = \int_0^{\infty} s_1(x)\,f(x)dx = 0.$$

- Similarly for the second term,

$$\mathbb{E}\left[s_1(X_i)\cdot \theta\right] = \theta\int_{-\infty}^{\infty} s_1(x)\,f(x)dx = \theta\int_0^{\infty} s_1(x)\,f(x)dx = 0.$$

- Hence the projection of the score for θ on the tangent set is $s(x) = 0$.
- Therefore, the efficient score and influence function are

$$\psi(x) = \mathcal{S}_\theta(x) = \frac{\mathbf{1}_{x>0} - \theta}{\theta(1-\theta)}, \qquad \text{and} \;\; \varphi(x) = \mathbf{1}_{x>0} - \theta,$$

and the semiparametric variance bound is (as we knew already)

$$\mathbb{E}[\varphi(X_i)\varphi(X_i)^\top] = \theta\cdot(1-\theta)$$

A GENERALIZATION

- Now this was a special case where the parameter of interest, θ, was actually directly a parameter of the parametric submodel.
- That need not be the case.
- For the average treatment effect we have the nuisance functions

$$\mu_w(x) = \mathbb{E}[Y_i | X_i = x, W_i = w], \qquad e(x) = \mathbb{E}[W_i | X_i = x],$$

and

$$\tau = \mathbb{E}\left[\mu_1(X_i) - \mu_0(X_i)\right].$$

- How can we find the efficiency bound there?

A GENERALIZATION

- Suppose the density of the random variable is $f(x)$, and the parameter θ is some functional of this density.
- Consider a parametric model for the density,

$$f(x;\gamma),$$

- score

$$\mathcal{S}(x;\gamma) = \frac{\partial \ln f}{\partial \gamma}(x;\gamma).$$

- In the parametric model θ can be written as a function of γ, $\theta = \theta(\gamma)$.

A GENERALIZATION

- What is the variance bound for γ for this parametric submodel?
- The Cramér-Rao variance bound for γ is

$$\mathrm{ASV}(\hat{\gamma}^{\mathrm{mle}}) = \left(\mathbb{E}\left[\mathcal{S}(Z_i;\gamma)\mathcal{S}(Z_i;\gamma)^\top\right]\right)^{-1}$$

- What is the variance bound for $\theta = \theta(\gamma)$?
- Use the delta method:

$$\mathrm{ASV}(\hat{\theta}^{\mathrm{mle}}) = \frac{\partial\theta}{\partial\gamma^\top}(\gamma)\left(\mathbb{E}\left[\mathcal{S}(Z_i;\gamma)\mathcal{S}(Z_i;\gamma)^\top\right]\right)^{-1}\frac{\partial\theta^\top}{\partial\gamma}(\gamma)$$

A GENERALIZATION

- Suppose there is a function $\phi(\cdot)$ (the Riesz representer) such that

$$\frac{\partial\theta}{\partial\gamma}(\gamma) = \mathbb{E}\left[\phi(Z_i)^\top \mathcal{S}(Z_i;\gamma)\right]$$

  for all smooth parametric submodels.

- The existence of the Riesz representer is a property of the parameter θ, referred to as differentiability of the parameter θ.
- So, we are only looking at efficiency bounds for differentiable parameters.

A GENERALIZATION

- Let us look at $\theta = \mathbb{E}[Z_i]$:

$$\theta(\gamma) = \mathbb{E}[Z_i] = \int z\, f(z;\gamma)dz$$

- Then:

$$\frac{\partial\theta}{\partial\gamma}(\gamma) = \int z\frac{\partial\, f}{\partial\gamma}(z;\gamma)dz = \int z\frac{\partial \ln\, f}{\partial\gamma}(z;\gamma)\, f(z;\gamma)dz$$

$$= \int z\mathcal{S}(z;\gamma)\, f(z;\gamma)dz = \mathbb{E}[Z_i\mathcal{S}(Z_i;\gamma)]$$

hence the Riesz representer is $\phi(z) = z$, and θ is differentiable.

A GENERALIZATION

- If differentiability holds, then the Cram'er-Rao variance bound for $\theta = \theta(\gamma)$ (in the parametric case) can be characterized in terms of the Riesz representer as

$$\mathbb{E}\left[\phi(Z_i)^\top \mathcal{S}(Z_i;\gamma)\right]\left(\mathbb{E}\left[\mathcal{S}(Z_i;\gamma)\mathcal{S}(Z_i;\gamma)^\top\right]\right)^{-1}\mathbb{E}\left[\mathcal{S}(Z_i;\gamma)^\top \phi(Z_i)\right]$$

$$= \mathbb{E}\left[\varphi(Z_i;\gamma)\varphi(Z_i;\gamma)^\top\right]$$

- Here the influence function is the projection of the Riesz representer $\phi(Z_i)$ on the full score function $\mathcal{S}(Z_i,\gamma)$:

$$\varphi(z;\gamma) = \mathcal{S}(z_i;\gamma)^\top\left(\mathbb{E}\left[\mathcal{S}(Z_i;\gamma)\mathcal{S}(Z_i;\gamma)^\top\right]\right)^{-1}\mathbb{E}\left[\mathcal{S}(Z_i;\gamma)^\top \phi(Z_i)\right]$$

A GENERALIZATION

- Back to the semiparametric case where we have have a lot of score functions corresponding to different parametric submodels.
- Find the Riesz representer.
- Consider the tangent space based on the linear combinations of all possible score functions $\mathcal{S}(z;\gamma)$, which is the set of functions

$$\mathcal{T} = \left\{ s(\cdot) \,\middle|\, \int_x s(x)\, f(x)dx = 0 \right\}.$$

- Now project the Riesz representer $\phi(\cdot)$ onto this tangent space (here things get technical with projections in infinite dimensional spaces, you can look at the references for this).
- The projection $\varphi(\cdot)$ is the influence function.
- The semi-parametric variance bound is

$$\mathbb{E}[\varphi(X_i)\varphi(X_i)^\top]$$

A GENERALIZATION

- First, let us first do this for the simple example we have looked at earlier.
- Given a parametric model $f(x;\gamma)$, with score $\mathcal{S}(x;\gamma)$, we can write the parameter of interest as

$$\theta(\gamma) = \int_{-\infty}^{\infty} \mathbf{1}_{x>0} f(x;\gamma)dx,$$

- The derivative is

$$\frac{\partial\theta}{\partial\gamma}(\gamma) = \int_{-\infty}^{\infty} \mathbf{1}_{x>0} \frac{\partial f}{\partial\gamma}(x;\gamma)dx = \int_{-\infty}^{\infty} \mathbf{1}_{x>0} \mathcal{S}(x;\gamma) f(x;\gamma)dx$$

$$= \mathbb{E}\left[\left(\mathbf{1}_{X>0} - \int_{0}^{\infty} f(z)dz\right) \mathcal{S}(X;\gamma)\right].$$

- So we have the Riesz representer

$$\phi(x) = \mathbf{1}_{x>0} - \int_{0}^{\infty} f(z)dz,$$

A GENERALIZATION

- The tangent set is

$$\mathcal{T} = \left\{ s(\cdot) \,\middle|\, \int_{-\infty}^{\infty} s(x) f(x) dx = 0 \right\}.$$

- The Riesz representer $\phi(x) = \mathbf{1}_{x>0} - \int_0^\infty f(z)dz$ is an element of the tangent set because

$$\int_{-\infty}^{\infty} \phi(x) f(x) dx = 0.$$

- Hence the influence function, defined as the projection of the Riesz representer $\phi(\cdot)$ on the tangent set, is equal to the Riesz representer itself, $\varphi(\cdot) = \phi(\cdot)$
- The variance bound is

$$\mathbb{E}[\varphi(X_i)\varphi(X_i)] = \theta(1 - \theta).$$

THE AVERAGE TREATMENT EFFECT CASE

- Now let us consider the more complicated average treatment effect example.
- The joint density of (Y_i, X_i, W_i) can be written as

$$f(y,w,x) = f(y|x,w=1;\gamma)^{w} f(y|x,w=0;\gamma)^{(1-w)}$$

$$e(x;\gamma)^{w}(1-e(x;\gamma))^{(1-w)} f(x;\gamma).$$

- The parameter of interest is

$$\theta(\gamma) = \int\int y f(y|x,w=1;\gamma) dy dx - \int\int y f(y|x,w=0;\gamma) dy dx,$$

THE AVERAGE TREATMENT EFFECT CASE

- The full score function is

$$\mathcal{S}(y, w, x; \gamma) = w\mathcal{S}_1(y|x; \gamma) + (1-w)\mathcal{S}_0(y|x; \gamma) + \frac{w - e(x; \gamma)}{e(x; \gamma)(1 - e(x; \gamma))} e'(x; \gamma) + \mathcal{S}_x(x; \gamma),$$

- Here

$$\mathcal{S}_0(y|x; \gamma) = \frac{\partial \ln f}{\partial \gamma}(y|x, w = 0; \gamma),$$

$$\mathcal{S}_1(y|x; \gamma) = \frac{\partial \ln f}{\partial \gamma}(y|x, w = 1; \gamma),$$

$$e'(x; \gamma) = \frac{\partial e}{\partial \gamma}(x; \gamma), \qquad \mathcal{S}_x(y|x; \gamma) = \frac{\partial \ln f}{\partial \gamma}(x; \gamma).$$

THE AVERAGE TREATMENT EFFECT CASE

- The tangent set is the set of all linear combinations of all possible score functions

$$\mathcal{T} = \left\{ s(y, w, x) : \right.$$

$$\left. s(y, w, x) = w s_1(y|x) + (1 - w) s_0(y|x) + a(x)(w - e(x)) + s_x(x) \right\}$$

Here we have three restrictions that the four components $s_0(\cdot)$, $s_1(\cdot)$, $s_x(\cdot)$ and $a(\cdot)$ must satisfy:

- $s_1(y|x)$ satisfies $\int s_1(y|x) f(y|x, w = 1) dy = 0$,
- $s_0(y|x)$ satisfies $\int s_0(y|x) f(y|x, w = 0) dy = 0$,
- $s_x(x)$ satisfies $\int s_x(x) f(x) dx = 0$,
- and $a(x)$ can be anything.

THE AVERAGE TREATMENT EFFECT CASE

- The derivative of the parameter of interest with respect to the parameter of the parametric submodel is

$$\frac{\partial\theta}{\partial\gamma}(\gamma) = \int\int y\mathcal{S}_1(y|x;\gamma)\,f(y|x,w=1;\gamma)\,f(x;\gamma)dydx$$

$$+\int\int y\,f(y|x,w=1;\gamma)\mathcal{S}_x(x;\gamma)\,f(x;\gamma)dydx$$

$$+\int\int y\mathcal{S}_0(y|x;\gamma)\,f(y|x,w=0;\gamma)\,f(x;\gamma)dydx$$

$$+\int\int y\,f(y|x,w=0;\gamma)\mathcal{S}_x(x;\gamma)\,f(x;\gamma)dydx$$

THE AVERAGE TREATMENT EFFECT CASE

- Now guess for the influence function

$$\varphi(y,w,x) = w\frac{y - \int_y y f(y|x,w=1)dy}{e(x)} - (1-w)\frac{y - \int_y y f(y|x,w=0)dy}{1-e(x)}$$

$$+ \left(\int_y y f(y|x,w=1)dy - \int_y y f(y|x,w=0)dy - \theta\right)$$

$$= w\frac{y-\mu_1(x)}{e(x)} - (1-w)\frac{y-\mu_0(x)}{1-e(x)} + \left(\mu_1(x) - \mu_0(x) - \theta\right)$$

- This choice works to show that θ is a differentiable parameter, because

$$\frac{\partial\theta}{\partial\gamma}(\gamma) = \mathbb{E}\left[\varphi(Y_i,W_i,X_i)\mathcal{S}(Y_i,W_i,X_i;\gamma)\right].$$

- So the Riesz representer $\phi(Y_i,W_i,X_i)$ is equal to the influence function $\varphi(Y_i,W_i,X_i)$

THE AVERAGE TREATMENT EFFECT CASE

- In addition, the guess for the influence function $\varphi(y, w, x)$ is an element of the tangent set, so equal to its own projection, and therefore it is in fact equal to the influence function, and the variance bound is

$$\mathbb{E}[\varphi(Y_i, W_i, X_i)\varphi(Y_i, W_i, X_i)]$$

$$= \mathbb{E}\left[\left(W_i \frac{Y_i - \mu_1(X_i)}{e(X_i)} - (1 - W_i)\frac{Y_i - \mu_0(X_i)}{1 - e(X_i)} + \left(\mu_1(X_i) - \mu_0(X_i) - \theta\right)\right)^2\right]$$

$$= \mathbb{E}\left[\frac{\sigma_1^2(x)}{e(X_i)} - \frac{\sigma_0^2(X_i)}{1 - e(X_i)} + \left(\mu_1(X_i) - \mu_0(X_i) - \theta\right)^2\right].$$

- Bottom line: we can look for an estimator $\hat{\tau}$ that satisfies

$$\sqrt{N}\,(\hat{\tau} - \tau) \xrightarrow{d} \mathcal{N}\left(0, \mathbb{E}\left[\frac{\sigma_1^2(x)}{e(X_i)} - \frac{\sigma_0^2(X_i)}{1 - e(X_i)} + \left(\mu_1(X_i) - \mu_0(X_i) - \theta\right)^2\right]\right)$$

and we would know for sure that we have an efficient estimator.

CONSTANT TREATMENT EFFECT CASE

- Here is a second example. Suppose we have a completely randomized experiment:

$$W \perp\!\!\!\perp \Big(Y_i(\mathrm{C}), Y_i(\mathrm{T}) \Big),$$

- Suppose $p = \mathrm{pr}(W_i = 1)$,
- We are interested in the average treatment effect,

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$$

- The variance bound is

$$\mathbb{V} = \frac{\sigma_T^2}{p} + \frac{\sigma_C^2}{p} \qquad \text{withestimator} \hat{\tau} = \overline{Y}_T - \overline{Y}_C$$

The variance can be very big with thicktailed distributions.

- What if we know $Y_i(1) = Y_i(0) + \tau$? How does that change the variance bound?

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